Feature Analysis for Effective Tropical Cyclone Forecasting: Evaluating the Relevance of Data Inputs

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ABSTRACT

Accurate forecasting of tropical cyclones is crucial for minimizing the impact of these devastating natural phenomena. This study focuses on feature analysis to enhance the effectiveness of tropical cyclone forecasting models. By evaluating the relevance of various data inputs, including but not limited to latitude, longitude, wind speed, direction, and atmospheric pressure, this research aims to identify the most significant predictors for precise cyclone trajectory predictions. Through rigorous analysis and validation, this research aims to determine which data inputs are most critical for enhancing the reliability and accuracy of tropical cyclone forecasting. The findings will contribute valuable insights into feature selection strategies, aiding in the development of more effective and robust forecasting systems for disaster management and emergency response.

Keywords*: Forecasting, Cyclone Landfall Forecasting,*

1. INTRODUCTION

Tropical cyclones, also known as typhoons or hurricanes, are extremely harsh weather phenomenon. Between 1998 and 2017, storms, including those of tropical cyclones and hurricanes, came second only to earthquakes causing fatalities. These killed 233,000 people and during the same period injured, made homeless, displaced, or evacuated about 726 million people worldwide during the emergency phase of the disaster. Over the last 30 years, the number of people living on coasts exposed to cyclones rose by 192%, raising mortality and morbidity in the case of a tropical cyclone [1].

The geographic location of the Philippines makes it highly vulnerable to tropical cyclones, which cause extensive damage and loss of life, particularly in regions like the Cagayan Valley [2]. Understanding and predicting the path of tropical storms could be of great help in enhancing disaster preparedness, risk mitigation efforts, and emergency response strategies in vulnerable coastal regions.

1.1 Review of Related Literature

Previous studies have demonstrated the potential of deep learning models in improving the accuracy of tropical cyclone track forecasts. In the study "Forecasting tropical cyclone tracks in the Northwestern Pacific based on a deep-learning model" by Wang et al.(2022), they estimated the application of deep learning technology for the TC track forecasting in the Northwestern Pacific from 1979 to 2021. The data were divided into training (1979–2014), validation (2015–2018), and test sets (2019–2021), in which 6–72 h TC track forecasts were created using the historical trajectory data. They compared the performance of the three recurrent neural networks, namely RNN, LSTM, and GRU, and found that the best model was the GRU. They then proposed a combined model of GRU and convolutional neural network named GRU_CNN in further enhancement to the accuracy of the forecasts, integrating reanalysis data on the steering flow, sea surface temperature, and geopotential height. They utilized a combination of recurrent neural networks (RNN, LSTM, GRU) and proposed a GRU-CNN model that integrated reanalysis data on steering flow, sea surface temperatures, and geopotential height. This model outperformed traditional methods, particularly in short-term forecasts, demonstrating significant improvements in prediction accuracy[4].

Another approach is given by Tamamadin et al. (2022) in their paper, "Regional Typhoon Track Prediction Using Ensemble k-Nearest Neighbor Machine Learning in the GIS Environment." The researchers developed an ensemble k-nearest neighbor machine learning method for predicting typhoon tracks. The fact that each past track is divided into classes made the supervised classification/regression approach give very high accuracy and very short simulation times. Their approach indicated that the measurement of the similarity between historical typhoons at an early stage of the forecast process helps boost the prediction's accuracy [5].

In this study, we focus on employing the Long Short-Term (LSTM) algorithm to forecast the movement of tropical cyclones in the Philippines. The aim is to evaluate the effectiveness of LSTM in predicting cyclone trajectories and compare the results with another dataset commonly used in cyclone forecasting in the region. By analyzing the attributes of both datasets and identifying any disparities in forecasted outcomes, this research seeks to determine the reliability of different forecasting approaches. Such insights are crucial for enhancing the resilience of vulnerable communities and guiding decision-making processes during cyclone events.

The research aims to understand the importance of the variables in processing, analyzing, and providing accurate results for the landfall of a cyclone. The researchers divide the process into three parts: **(1)** Collect data such as acquiring multiple cyclone datasets for variable comparison purposes. **(2)** The data will then be pre-processed and analyzed through a Python notebook. **(3)** The processed data will then be presented such as the findings in this research.

2. METHODOLOGY

The methodology section provides an outline of a systematic approach applied to address the purpose of the research. It provides a detailed description of the steps taken to collect, analyze, and interpret the data to achieve the research objectives. In the context of predicting cyclone landfall based on datasets, this section will serve as a roadmap for conducting the study and providing knowledge-based insights

2.1 Data Collection

The researchers utilized the dataset extracted from the RSMC Japan Meteorological Agency. The dataset became the basis of comparison of the other datasets collected by the researchers. There are two datasets acquired for the comparison, namely: the Pacific, and historical cyclone datasets from Kaggle.

2.1.1 RSMC Dataset

For data collection, we will utilize the dataset available on the Japan Meteorological Agency website [3]. This dataset consists of various attributes, including indicators, international number IDs, the number of data lines, tropical cyclone number IDs, flags of the last data lines, and the time difference between the last data and the final analysis (measured in hours). It also provides information such as the name of the storm, the date of the latest revision, and the time of analysis in UTC format. The dataset categorizes cyclones based on their grades, which range from 1 (not used) to 9 (tropical cyclones of TS intensity or higher). Other attributes include the latitude and longitude of the cyclone center, central pressure (measured in hPa), maximum sustained wind speed (in knots), directions, and radii of 50 kt and 30 kt winds or greater, and indicators of landfall or passage. The dataset has already been simplified into numerical values on the website for ease of use.

The Regional Specialized Meteorological Centre (RSMC) Best Track Data is a standardized format utilized to record and store information about tropical cyclones. The data is vital for meteorologists, researchers, and policymakers to further enrich their knowledge on comprehending the behavior of the cyclones. The RSMC Tracking data usually includes (1) Storm Identification, (2) Data and Time, (3) Geographical Location, (4) Maximum Sustained Wind Speed, (5) Minimum Central Pressure, and (5) Radius of Maximum Winds (optional).

2.1.2 Pacific Dataset

The dataset utilized was taken from the National Oceanic and Atmospheric Administration (NOAA) and is available on Kaggle. The dataset contains attributes such as (1) ID, (2) date, (3) time, (4) status, coordinates such as (5) latitude and (6) longitude, and (7) maximum wind speed. The dataset contains data spanning several decades, providing historical context that is crucial for training machine learning models like the LTSM.

2.1.3 Indian Cyclones Dataset

The dataset utilizes multiple variables that are deemed to be beneficial in terms of forecasting the landfall of the cyclone. The dataset therefore specifically used (1) Basin of Origin, (2) Name of Cyclone, (3) (4) Date, (5) Time, (6) Year, (7) Latitude, (8) Longitude, (9) CiNo, (10) Estimated Central Pressure, (11) Maximum Surface Sustained Wind, (12) Pressure Drop, (13) Grade, (14) OCI, and (15) Diam. These variables that the dataset acquired, provide historical context that is aligned with the forecasting.

2.2 Data Preprocessing

The key steps in the model creation process include data preparation, sequence creation, and the design and training of the LSTM network. To analyze the trajectory/paths and model them effectively, it is necessary to preprocess and organize the data appropriately.

For the Pacific dataset, the researchers removed the name and minimum pressure attributes. Spaces from any string values are then removed from the dataset to ensure consistency. Numerical values are then mapped to different statuses in the “Status” attribute which contains categorical labels. Non-numeric

characters from Latitude and Longitude attributes are removed and then converted to float. Duplicate rows based on ID, Latitude, and Longitude are removed to represent a unique record for a specific tropical cyclone at a specific location. For the ID attribute, the first two letters for the labels are removed and the remaining string is converted to an integer to standardize the format in the ID attribute. Tropical cyclones with less than or equal to 20 data points for data consistency. Differences in latitude and longitude between consecutive data points within each cyclone track are computed by subtracting the latitude and longitude values of each data point from the previous data point, resulting in changes in latitude and longitude over time. The changes are then stored in the 'Delta Latitude' and 'Delta Longitude' columns. The arctangent of the ratio of the changes in longitude and latitude is calculated, which gives the movement direction in radians and then converted to degrees. The calculated degrees are then converted to compass directions (e.g., 'N', 'NE, 'W', etc.).

The dataset was loaded and certain attributes, such as 'Grade', were mapped to numerical values using a predefined dictionary. Duplicate records were removed to ensure data integrity.

TimeStamp attribute was converted to a datetime format and subsequently to a numerical representation (seconds since a reference point) to facilitate sequential modeling.

The data was grouped by 'TC_ID' to maintain the integrity of each tropical cyclone's sequence.

Additionally, the landfall cyclone prediction basing in the Indian historical data provides vivid representations of the predicted and actual landfall analysis wherein the landfall provides more accurate forecasting if there are more accessible variables that can be manipulated by the researchers.

During the pre-processing landfall, the dataset containing the historical data of cyclones from 1982 to 2020 is downloaded from the IMD website. The data is manually observed by the researchers and therefore the errors are corrected by applying standardization like dropping unnecessary variables. The dataset was parsed through a code to extract the features like Basin of Origin, Name of Cyclone which is optional, Date, Time, year, latitude, longitude, CiNo (T.No), Estimated Central Pressure (ECP), Maximum Surface Sustained Wind (MSSW), Pressure Drop (PD), Grade, OCI, and diam. The researchers only focused on the ECP, MSSW, latitude, and longitude features from the dataset.

The processing of data imputation the rows and columns that are empty. The imputation of newly inserted timestamp is done for filling the missing data that is filled per the process. The dataset was therefore merged with the Sea Surface Temperature (SST) data for NIO region, the dataset was downloaded from NOAA websites wherein the two datasets were merged using the model merging IMDSSTLandfall. The researchers therefore fetch the SST for the required locations in NIO. The resulting dataset was then exported as another csv. Lastly, the feature generation using the data generated at the last step of the process, researchers then generate two more features which is the distance and the direction between two consecutive recordings of a dataset to capture speed and change of angle. The next process of the dataset is to keep the recordings of a cyclone from the time it originates on the until it hit the lands.

2.3 Models and Structures

The models were trained using the Python programming language in Jupyterlab. Various libraries were utilized such as matplotlib, pandas, numpy, and folium. The said libraries were deemed necessary as the tropical cyclone prediction analysis will be based on the predicted values. Folium libraries were deemed vital, as the numerical values predicted and actual can be plotted in the real-life world map. Thus providing a quality visualization of the plotters.

Each dataset will then be run through this model, to check the accuracy of the landfall prediction for the past and upcoming tropical cyclones.

3. RESULTS AND DISCUSSION

The results and discussion section provides support to the findings and therefore provides a systematic method of applying the trained model. The research utilized two methods and a set of cyclone datasets to compare and analyze the importance of the variables in identifying and predicting the accuracy of the landfall of a cyclone.

Figure 1.1 Landfall Intensity from Start to Finish

In figure 1.1, the graph provides visuals of the predicted and actual data. During the analysis the researchers contributed thoroughly to the predictive analysis of cyclone landfall forecasting.

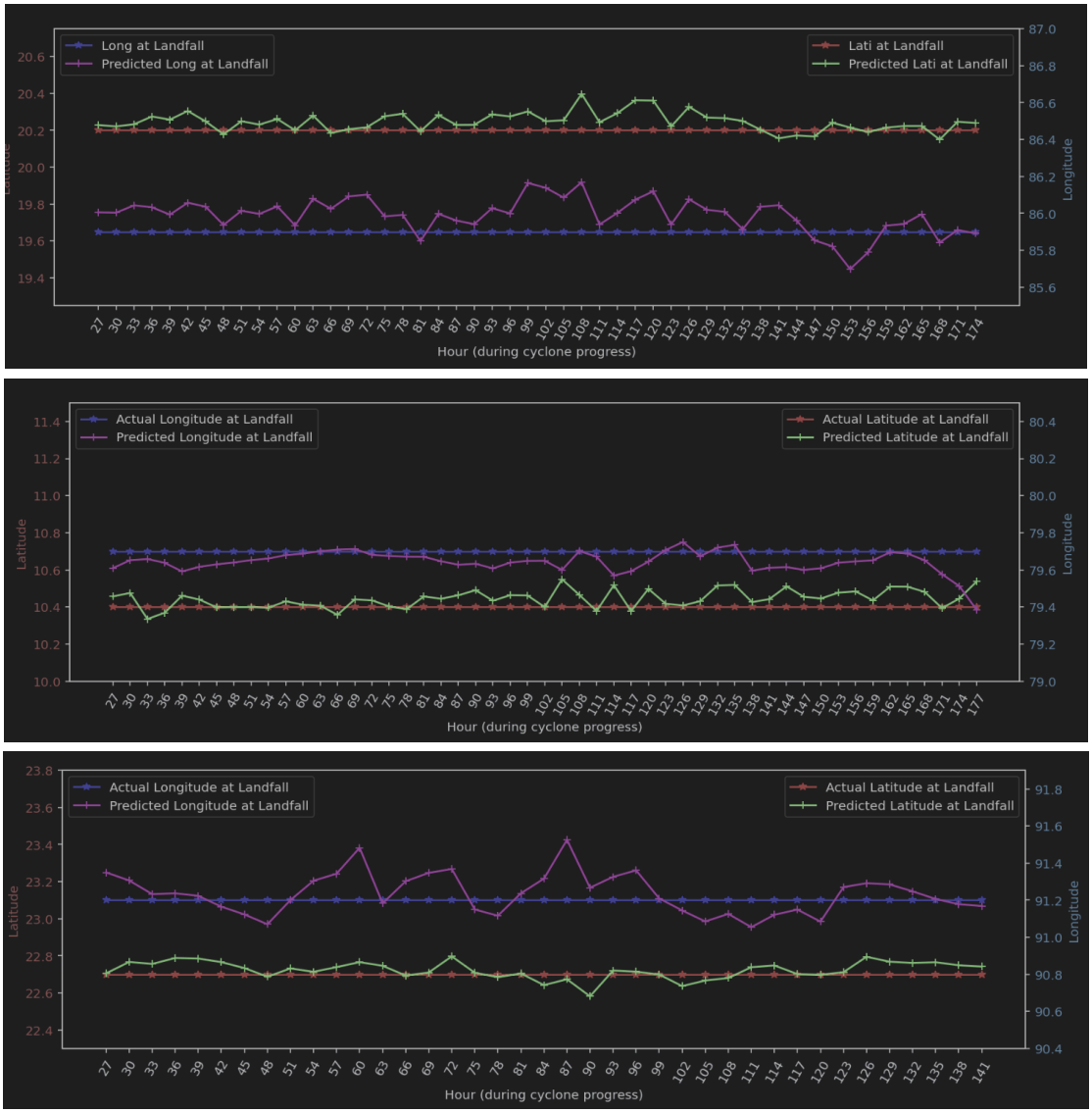
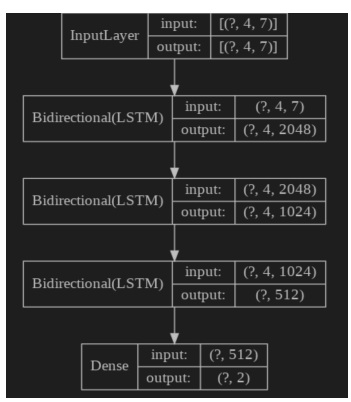


Figure 1.2 Latitude, Longitude and Landfall from Start Distance

Figure 1.2 provides the predicted and actual data of the landfall, therefore in the visualization the top three tropical cyclones.



3.1 Data Analysis

In time series forecasting, LSTM networks and ARIMA models are commonly used. However, there are certain justifiable reasons to prefer LSTM over ARIMA, especially for the task of tropical cyclone track prediction.

One such study was done by Salman et al., which compared the performance of the LSTM and ARIMA models in weather forecasting. They discovered that the ARIMA was effective for a few applications, while in most cases it does not embed intermediate variables in the model and is not able to capture complex relationships within the data. Their study proved that if the ARIMA model was used for the weather visibility forecast, the RMSE would give a value of 0.948 [6].

On the other hand, the LSTM model, which enabled the use of intermediate variables of temperature, humidity, and dew point, significantly outperformed ARIMA and achieved an RMSE value as low as 0.00007. This large improvement attests that LSTM, with respect to a study by Salman et al., is superior when working with multivariate data and eliciting the underlying nonlinearity in weather patterns.

This makes the LSTM model, incorporating several variables, quite promising for the prediction of tropical cyclone tracks and a mixture of SST, atmospheric pressure, and wind patterns. Indeed, the LSTM model estimates much better and more reliably than ARIMA, where, within the frame of univariate analysis, it risks overlooking many important dependencies. Among these advantages, we considered LSTM as appropriate for our study to predict tropical cyclones' movement in the Philippines. It is the approach that will utilize LSTM's handling of complex multivariate time series data well and therefore improve the accuracy and reliability of the cyclone path predictions. Lastly, LSTM is more suitable for the type of dataset format we have.

3.2. Model Creation

RSMC MODEL

The model developed for this project is based on the Long Short-Term Memory (LSTM) algorithm, a type of Recurrent Neural Network (RNN) that is particularly useful for sequential data and time series forecasting.

To model the trajectories of tropical cyclones effectively, the researchers prepared the data in a way that the model can learn from historical patterns and future paths. This involved data sorting, grouping cyclone IDs, and finally creating sequences for training.

The sequence creation involves determining a fixed input sequence length and an output sequence length; this helps in creating uniform sequences that the LSTM model can process.

Key features were selected like ‘Latitude’, ‘Longitude’, ‘Grade’, ‘Direction Degrees’, and ‘Central Pressure’ for the sequences. Normalization was done before feeding the data using ‘MinMaxScaler’ to ensure all features are on a similar scale which helps in faster model training. For each position in the data, a sequence of length ‘input_seq_length’ was extracted as the input along with the output, ‘output_seq_length’.

‘GroupShuffleSplit’ was used to divide the data into training, testing, and unseen datasets. This ensures that the data is split based on ‘TC_ID’.

INDIAN CYCLONES MODEL

The Indian cyclone model utilizes a similar algorithm like the RSMC model which is the Long Short-Term Memory (LSTM). It is a type of recurrent neural network aimed at dealing with the vanishing gradient problem present in traditional RNNs. The model considered the latitude, longitude, SST, MSSW and ECP of the cyclone to be forecasted and predicted fully by the model. Just like in RSMC, the model also has a fixed input of length and output sequence lengths, these are due to the selection process done by the researchers wherein the top three (3) strongest cyclones are the ones that have been considered. The training, testing and validation of the dataset were done by splitting the variables.

3.3 Model Evaluation

RSMC MODEL

In this section, the researchers evaluate the performance of the LSTM model, which predicts the next 10 coordinates of a cyclone based on historical data.

For the first dataset (RMSC_DATASET), the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are calculated for both scaled and unscaled coordinates.

For the Scaled Metrics, the relatively low values (MAE: 0.2951, RMSE: 0.4174) indicate that the model's predictions are quite close to the actual scaled coordinates. The MAE result suggests that on average, the model's prediction deviates from the actual values by approximately 0.2951 units of the scaled coordinates. An RMSE of 0.4174 implies that on average, the prediction error is 0.4174 units in the scaled feature space.

For the Unscaled Metrics, a Mean Absolute Error of 6.7462 suggests that on average, the model's prediction deviates from the actual values by approximately 6.7462 units of the original coordinates. A RMSE value of 11.4355 implies that, on average, the prediction error is 11.4355 units in the original feature space. The higher values compared to the scaled metrics reflect the actual geographical errors in predicting cyclone coordinates.

INDIAN

The Indian cyclone forecasting model aims to predict the landfall time and location as close to the actual data as possible. The same with the RSMC cyclone forecasting model, this forecasting program utilizes the presence of Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

The program aims to identify the RMSE and MAE of the three cyclones which are Gaja, Fani, and Bulbul. The RMSE and MAE per cyclone in terms of its landfall, and time of arrival has been trained in the model and therefore the analysis of the program was plotted using the matplot.lib of python programming language which provides vivid and easy-to-interpret output of the linear regression.

RSMC DATASET

RSMC ATTRIBUTE GROUP 1

Features	Predict
● Grade ● year ● month ● day ● hour ● Latitude of the center ● Longitude of the center ● Central pressure ● Maximum sustained wind speed ● The longest radius of 50kt winds or greater ● The shortest radius of 50kt winds or greater ● The longest radius of 30kt winds or greater ● The shortest radius of 30kt winds or greater	● Latitude of the center ● Longitude of the center ● Central pressure ● Maximum sustained wind speed ● The longest radius of 50kt winds or greater ● The shortest radius of 50kt winds or greater ● The longest radius of 30kt winds or greater ● The shortest radius of 30kt winds or greater

Training

	Training Score	**Test Score**
Linear Regression	10.651949419322946	10.774686433737937
Random Forest	4.6626896065254	12.493161608821167
Deep Neural Network (MLP)	Epoch: 1 cost: 16898.345703125 Epoch: 2 cost: 722.1717529296875 Epoch: 3 cost: 642.1868896484375 Epoch: 4 cost: 526.2147216796875 Epoch: 5 cost: 496.5829162597656 Epoch: 6 cost: 431.5306396484375 Epoch: 7 cost: 403.2513427734375 Epoch: 8 cost: 359.2433776855469 Epoch: 9 cost: 333.16912841796875 Epoch: 10 cost: 329.23187255859375 Epoch: 11 cost: 302.8529357910156 Epoch: 12 cost: 290.28741455078125 Epoch: 13 cost: 276.1322326660156 Epoch: 14 cost: 269.1590270996094 Epoch: 15 cost: 256.7517395019531	13.78868658385819

Evaluation with 2022-2023 data

	Test Score
Linear Regression	9.360596359530215

Random Forest	13.038600367198953
Deep Neural Network (MLP)	12.477604417270518

RSMC ATTRIBUTE GROUP 2

Features	Feed	Predict
● Latitude of the center ● Longitude of the center ● Central pressure ● Maximum sustained wind speed ● The longest radius of 50kt winds or greater ● The shortest radius of 50kt winds or greater ● The longest radius of 30kt winds or greater ● The shortest radius of 30kt winds or greater	● year ● month ● day ● hour ● Grade ● Latitude of the center ● Longitude of the center ● Central pressure ● Delta Latitude ● Delta Longitude ● Direction Degrees	● Grade ● Latitude of the center ● Longitude of the center ● Central pressure ● Delta Latitude ● Delta Longitude ● Direction Degrees

Training

	Training Score	**Test Score**
Linear Regression	5.433509921811769	5.371154941950325
Random Forest	2.291325554212222	6.0776608456025345
Deep Neural Network (MLP)	5.61941143487065	Epoch: 1 cost: 11203.099609375 Epoch: 2 cost: 194.70803833007812 Epoch: 3 cost: 165.63265991210938 Epoch: 4 cost: 160.7236328125 Epoch: 5 cost: 159.1031036376953 Epoch: 6 cost: 158.3679962158203 Epoch: 7 cost: 156.56239318847656 Epoch: 8 cost: 156.61453247070312 Epoch: 9 cost: 155.80691528320312 Epoch: 10 cost: 156.65777587890625 Epoch: 11 cost: 156.14283752441406 Epoch: 12 cost: 156.68649291992188 Epoch: 13 cost: 155.5478057861328 Epoch: 14 cost: 155.92733764648438 Epoch: 15 cost: 155.9856414794922

Evaluation with 2022-2023 data

	Test Score
Linear Regression	4.770858260903174
Random Forest	6.400568331204359
Deep Neural Network (MLP)	4.995516463336894

INDIAN CYCLONE MODEL

In this section, the researchers also assessed the performance of the LSTM model, which predicts the cyclone coordinates of a cyclone based on historical data.

The Indian cyclone model utilizes the processingLandfall, mergingIMDSSTLandfall, featureGeneration, LatiLongAtLandfallFromStartDistance, and LandfallIntensityAndTimeFromStart wherein the dataset was analyzed. The LatiLongAtLandfallFromStartDistance contains the experimental tool to provide the results regarding Landfall's latitude, longitude, and distance error between predicted and actual landfall location. Graphs are shown by the program by the utilization and imported the data to another file.

Accurate prediction of the path of a typhoon is very important for making plans under a scenario that involves this kind of destructive natural catastrophe. This paper explores a Bidirectional Long Short-Term Memory (BiLSTM) model for forecasting the typhoon path using evaluation metrics like RMSE, MAE, and average geographical distance errors. Model performance was evaluated via cross-validation and particular cyclone evaluation for Fani, Gaja, and Bulbul.

Model Evaluation Metrics

The performance evaluation of the BiLSTM model was done using several metrics:

- Root Mean Squared Error (RMSE): The square root of the average of squared differences between the actual and predicted values.
- Mean Absolute Error (MAE): Measures the average absolute differences between predicted and actual values.
- Average Distance Error: Average geodesic distance between predicted and actual cyclone tracks, reported in both kilometers and miles.

These were calculated separately for latitude and longitude predictions in order that one can get full insight into the accuracy of the model.

Cross-Validation Results The model has been developed with strong robustness and reliability during 5-fold cross-validation. Table 1: Cross-Validation Performance of BiLSTM Model Table shows a summary of the average errors for both latitude and longitude predictions across the folds.

Cross-Validation Performance of BiLSTM Model Table

Metric	Score
Latitude RMSE	1.335
Longitude RMSE	2.327
Latitude MAE	0.997
Longitude MAE	1.654
Distance (Miles)	140.419
Distance (KM)	225.339

Model performance was tested further for individual cyclones: Fani, Gaja, and Bulbul. The detailed performance metrics of these cyclones are listed in the following Performance on Specific Cyclones Table.

Performance on Specific Cyclones

Cyclone	Latitude RMSE	Longitude RMSE	Latitude MAE	Longitude MAE	Distance (KM)	Distance (Miles)
Fani	1.035	1.414	0.870	1.153	164.517	102.256
Gaja	0.972	0.321	0.804	0.262	100.748	62.596
Bulbul	0.973	1.268	0.832	1.062	159.518	99.132

From cross-validation, it is possible to say that the BiLSTM model slightly performs better in the prediction of latitude, as shown by the lower RMSE and MAE values of latitude in comparison to the prediction of longitude. The average RMSE in the predictions of latitude is 1.335, while for longitude, it is 2.327. At the same time, the average MAE for latitude is 0.997, and for longitude, it is 1.654. These

values suggest that although the accuracy is rather high, the model can still be improved by further tuning toward the prediction of longitude.

It shows the best for Cyclone Gaja, with the least RMSE and MAE among the latitude and longitude predictions, on individual cyclone performance. The ones preferred by the errors in geographical distance are Gaja which shows an average error of distance of 100.748 km (62.596 miles), Fani, and Bulbul.

The potential of the BiLSTM model has been too high for the prediction of typhoon pathways, giving latitude. The evaluation at the time of cyclones Fani, Gaja, and Bulbul shows the possibility and the effectiveness of the method in operational practice and gives longitudes with the best precision in the case of Gaja. Still, there is a good scope of improvement. Hence, efforts can be made in the future for the improvement of the model and the addition of other data into the model for better predictions, leading in turn to better-prepared responses to typhoon impacts.

4. CONCLUSION

The countries surrounding the tropical areas remain the most susceptible to cyclones, which are a significant cause of threat to life and properties. With the population increasing day by day, a major portion of it resides closer than ever to the coasts. This is one way in which tropical cyclones pose a great threat to the Philippines, thus mandating a strong technique in forecasting to help alleviate such vulnerability. For this reason, there is an alarming need to create accurate and reliable cyclone forecasts.

We have used detailed datasets from both RSMC Japan Meteorological Agency and NOAA Pacific, provided on Kaggle. The data has been transformed, cleaned, and normalized at all preprocessing steps to make it model-ready. The output of the LSTM model is quite decent, showing relatively low MAE and RMSE pointing to very close alignment with the real coordinates of cyclones. This accuracy supports the potential operability of the model in disaster preparedness and response. Previous works have shown deep learning models, including LSTM, perform far better in improving the accuracy of cyclone forecasts than traditional methods.

This study tries to investigate if the Long Short-Term Memory (LSTM) algorithm can be efficient for tropical cyclone trajectory prediction based on a dataset from the Japan Meteorological Agency and the historical cyclone dataset published on Kaggle. We confirmed that in this study, where the LSTM model was successful with the complex multivariate time series data.

Additionally, in the Indian cyclone forecasting, we utilized and devised a model which is landfall intensity and time from start. The model was therefore utilized to provide visuals by plotting the data points in graphs and having the predicted and actual data to be presented simultaneously. Furthermore, the dataset that utilizes the dataset with more details provides more accurate landfall prediction. The latitude, longitude, and wind pressure that the Japan Cyclone dataset contains are deemed to be lacking thus considering the variables from the Indian Cyclone dataset contain more accessible information to be manipulated by the researchers and therefore forecasted a more accurate landfall prediction.

The implications of such a study include an early warning with the accurate forecasting of cyclone trajectories, which may further help make the appropriate arrangements for people's evacuation and better allocation of services and resources in vulnerable regions. Reliable forecasting models help in the reduction of the negative effects of tropical cyclones on communities. This study can thus be extended further by increasing prediction accuracy through the fusion of deep-learning models with other meteorological data sources. Continuous model updates with new data will also ensure the forecasting system remains effective.

In conclusion, integrating the potential of advanced deep learning algorithms with large meteorological data sets, such as LSTM, in-depth, holds great promise for improvements in tropical cyclone forecasting. This study contributes insights and knowledge into the development and application of models similar to this study.

REFERENCES

[1] World Health Organization. Tropical Cyclones. Retrieved May 15, 2024 from https://www.who.int/health-topics/tropical-cyclones

[2] Maribel S. Abalos and Arnel C. Fajardo. 2023. Tropical Cyclone Analysis and Accumulated Precipitation Predictive Model Using Regression Machine Learning Algorithm. In *Computer and Communication Engineering*, 2023. Springer Nature Switzerland, Cham, 203–219. https://doi.org/10.1007/978-3-031-35299-7_17

[3] Japan Meteorological Agency. RSMC Tokyo - Typhoon Center. Retrieved May 15, 2024. from https://www.jma.go.jp/jma/jma-eng/jma-center/rsmc-hp-pub-eg/RSMC_HP.html

[4] Wang, L., Wan, B., Zhou, S., Sun, H., and Gao, Z.: Forecasting tropical cyclone tracks in the northwestern Pacific based on a deep-learning model, Geosci. Model Dev., 16, 2167–2179, https://doi.org/10.5194/gmd-16-2167-2023, 2023.

[5] Tamamadin, M., Lee, C., Kee, S., and Yee. J. 2022. Regional Typhoon Track Prediction Using Ensemble k-Nearest Neighbor Machine Learning in the GIS Environment. Remote Sensing 14, 21 (2022), 5292. DOI:https://doi.org/10.3390/rs14215292.

[6] Salman, A., Heryadi, Y., Abdurahman, E., and Suparta, W. 2018. Weather Forecasting Using Merged Long Short-Term Memory Model (LSTM) and Autoregressive Integrated Moving Average (ARIMA) Model. Journal of Computer Science 14, 7 (2018), 930-938. DOI:https://doi.org/10.3844/jcssp.2018.930.938